

OSCAR OLIVA

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EDUCATION

University of Pittsburgh – Swanson School of Engineering

Graduated April 2025

- Bachelor of Science in Mechanical Engineering
- Minor in Bioengineering

PROJECTS / ACCOMPLISHMENTS

- Conductively Coated Fabric Electrode – Patent Pending | Co-Inventor (ZOLL)
- Engineering Portfolio is available via the QR code or at: <https://oaoliva.github.io/>

ENGINEERING EXPERIENCE

Wood

Pittsburgh, PA

Field Vibration Analyst

May 2025 - Present

- Visit client sites to collect vibration data on gas compressors, focusing on piping systems and small bore piping.
- Analyze vibration data, natural frequencies, and operating deflection shapes to identify possible failure modes.
- Develop targeted solutions for identified problem areas in comprehensive reports and present findings to stakeholders.

Human Movement and Balance Laboratory, Department of Bioengineering

Pittsburgh, PA

Research Assistant (Part-time during academic semesters)

November 2021 – April 2025

- Integrated a Raspberry Pi and camera into a portable scanner which used frustrated total internal reflection (FTIR) technology, enabling automated capture, storage, and analysis of tread wear images upon user interaction.
- Developed MATLAB-based image processing software to identify worn shoe regions from FTIR data and utilized Python to correlate image brightness with pressure exerted on the shoe.
- Adapted tribology testing equipment to evaluate friction on various ladder rungs under dry and contaminated conditions; designed custom supports and established a robust testing protocol.
- Analyzed tribology data in MATLAB to quantify relationships between surface patterns, materials, rung geometries, and coefficients of friction.

ZOLL LifeVest

Pittsburgh, PA

Human Factors Engineering Co-op (three 4-month rotations)

September 2022 – April 2024

- Conducted post-market surveillance analysis on an existing product, contributing to the HFE/UE validation report submitted to the FDA for the next-generation product.
- Developed protocols for wear test evaluations and moderated studies by interviewing participants to assess comfort, usability, and the effectiveness of error-prevention features of the LifeVest system.
- Leveraged R and Power BI to analyze study data, prepared detailed reports, presented findings, and collaborated with a multidisciplinary engineering team to address critical use errors.

Advanced Research / Advanced Development

May 2024 – July 2024

- Designed a prototype LifeVest garment to enhance patient comfort while maintaining ECG quality by incorporating fabric electrodes and relocating circuitry to the garment's exterior.
- Applied 3D scanning to capture patient body measurements for garment sizing and analyzed correlations between ECG wave amplitudes and electrode placement across the body.

Desapro Inc.

Rockledge, FL

Mechanical Engineering Intern

May 2022 – August 2022

- Drafted SolidWorks models of aluminum transit cases and prepared detailed production-ready drawings using GD&T.
- Analyzed sheet metal-fabricated cases for manufacturability and implemented design adjustments using SolidWorks, resulting in improved production efficiency.
- Collaborated with customers to deliver custom cases tailored to specific size, weight, and load-bearing requirements, leading both design and manufacturing efforts.
- Designed and managed the manufacturing of a vibration-dampening case using advanced materials to reduce impact shock in sensitive equipment.

SKILLS

Technical: SolidWorks, Fusion 360, MATLAB, Python, C, GD&T

Rapid Prototyping: Soldering, 3D Printing, 3D Scanning, Injection Molding, Instron Testing

Professional: Technical Writing, Presenting, Spanish (Fluent)